

Blood Oxygen and Heart Rate Monitor

Category: Health Monitoring | Status: Completed Prototype | 7 min read

A MAX30102-based Arduino project that explores how pulse oximetry sensors estimate heart rate and blood oxygen trends.

Materials

- Arduino UNO R3
- MAX30102 pulse oximetry sensor module
- LCD 1602 or OLED display
- Breadboard
- Jumper wires
- USB cable or battery source

What I learned

- Biomedical data starts as a physical signal.
- Filtering and stability matter before a reading becomes useful.
- Honest safety notes are essential for health-related projects.

Safety or accuracy note

This prototype is for learning and demonstration only. It is not a medical device and should not be used for diagnosis or treatment.